

Same network minima, different saddle spectra

Proof of conjecture 00000006080

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The conjecture asks for two networks with identical minimum positions and values but different saddle spectra, obtained by an explicit perturbation. We construct two genuine feedforward networks sharing both square-activation hidden layers. Changing one output weight preserves every local and global minimum but changes the Hessian spectrum of their common saddle.

Actual network architecture. The activation is $s(u) = u^2$ at each hidden neuron. For input $(x, y) \in \mathbb{R}^2$, the first affine layer and activation give

$$h_1 = (x^2, (y+1)^2, (y-1)^2).$$

The second affine layer uses weights

$$W_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/4 & -1/4 \end{pmatrix}, \quad b_2 = (-1, 0),$$

so its preactivation is $(x^2 - 1, ((y+1)^2 - (y-1)^2)/4) = (x^2 - 1, y)$. After squaring, the hidden output is $((x^2 - 1)^2, y^2)$. The final linear output weights $(c, 1)$ yield

$$L_c(x, y) = c(x^2 - 1)^2 + y^2.$$

The first layer has weights with rows $(1, 0)$, $(0, 1)$, $(0, 1)$ and bias $(0, 1, -1)$. Thus every affine layer, activation and output weight is specified. Our two networks are L_1 and L_2 . The source fixes no activation class; we use square activation.

Every minimum position and value. For every $c > 0$, $L_c \geq 0$, with equality exactly at

$$\mathcal{P} = \{(-1, 0), (1, 0)\}.$$

Indeed both summands are nonnegative; their sum vanishes exactly when $x^2 = 1$ and $y = 0$. Therefore these are precisely all global minimum positions, each with value zero.

To cover all local minima as well, the actual derivative is

$$DL_c(x, y)[u, v] = 4cx(x^2 - 1)u + 2yv.$$

Every local minimum is critical, hence $y = 0$ and $x \in \{-1, 0, 1\}$. The origin is not a local minimum: for arbitrarily small $0 < t < 1$,

$$L_c(t, 0) - L_c(0, 0) = ct^2(t^2 - 2) < 0.$$

The remaining two critical points are global minima. Consequently L_1 and L_2 have exactly the same local minima, global minima and minimum values.

Actual saddle Hessians and full spectra. The second derivatives give the actual Hessian at every point:

$$H_c(x, y) = \begin{pmatrix} 4c(3x^2 - 1) & 0 \\ 0 & 2 \end{pmatrix}.$$

The common critical point $(0, 0)$ has

$$H_c(0, 0) = \begin{pmatrix} -4c & 0 \\ 0 & 2 \end{pmatrix}.$$

For $c > 0$, one eigenvalue is negative and the other positive, so the origin is a strict saddle. Directly, small x -axis perturbations decrease the loss, while nonzero y -axis perturbations increase it.

The characteristic determinant is

$$\det(zI - H_c(0, 0)) = (z + 4c)(z - 2).$$

Since a complex square matrix is invertible exactly when its determinant is nonzero, this characterizes the entire actual spectrum:

$$\sigma(H_1(0, 0)) = \{-4, 2\}, \quad \sigma(H_2(0, 0)) = \{-8, 2\}.$$

These spectra differ. The origin is the only saddle critical point: the other two critical points are the global minima.

Explicit perturbation. Only the first output weight changes, from one to two:

$$L_2(x, y) - L_1(x, y) = (x^2 - 1)^2.$$

More generally the explicit output-weight path $c = 1 + \theta$, $0 \leq \theta \leq 1$, leaves every minimum position and value unchanged, while the saddle's negative eigenvalue moves from -4 to -8 . This realizes the requested separation.

Formal correspondence. Lean defines actual dense affine layers with weight matrices and biases, square activation and a feedforward network structure. `network_formula` derives the displayed polynomial from the actual layer evaluation, rather than declaring it a network.

The derivative theorem proves the full `HasFDerivAt` formula. The local-minimum theorem and global-minimum theorem prove the complete minimum-position sets for every $c > 0$, and `minimum_value` proves value zero. The local-minimum exclusion uses an actual arbitrarily close descent point in each neighborhood.

`Hessian` is defined by actual second partial derivatives of its function argument. The Hessian formula computes them, and the spectrum theorem proves the full Mathlib spectrum using invertibility and determinants. `StrictSaddle` requires a zero Fréchet derivative and negative and positive points in the actual Hessian spectrum. The final `counterexample` existentially supplies the two actual networks, their identical minima and values, their strict saddles, unequal spectra and explicit perturbation.

Reproduction. Run `lake build in lean/` using Lean 4.19.0 and public Mathlib revision

`c44e0c8ee63ca166450922a373c7409c5d26b00b`

The project prints axiom audits.